

Why PFAS half-lives differ between humans, rats and mice

A review of the evidence, the assumptions behind the published numbers, and what is still unresolved

Evidence base	186 full texts read; 24 per-paper extractions; 1,438 database rows; 51 references
Underlying review	report/REPORT.md, ~2,002 lines, 34 runnable scripts, 16 figures
Repository	github.com/sdey17/pfas_tk

Contents

1	Introduction: one identity, three questions
2	Evidence base and method
3	Where the species difference lives: distribution or clearance?
4	The mechanism: fractional renal reabsorption
5	Transporter identity, and why the mechanism does not transfer
6	The second loop: enterohepatic recirculation
7	Plasma protein binding, and a conflict that was not one
8	Does exposure level change elimination?
9	The human volume of distribution
10	Toward structure-activity: the renal handling ratio
11	What this review corrects in the published record
12	Coverage: how much of the field is empty
13	Open questions, and what would close them
14	Conclusions
	References

Abstract

Published serum half-lives for the same per- and polyfluoroalkyl substance differ by orders of magnitude between humans and laboratory animals, between rat and mouse, between male and female, and sometimes between papers describing the same experiment. This review asks where that variance actually sits. Because half-life is not a property but a ratio - the volume a chemical apparently occupies divided by the rate the body clears it - the question is answerable, and the answer is unambiguous: across six datasets that measure both terms in both sexes within single experiments, the volume of distribution varies 1.6-fold while clearance varies 56-fold and changes sign between species. The variance is in clearance.

One mechanistic axis then orders every species for which data exist. Fractional renal reabsorption runs from 99.94% in humans through 97% in the male mouse and 93.7% in the male rat to net tubular secretion in the female rat and the rabbit, and it reproduces rodent half-lives within 1.7-fold over a 37-fold span with no fitted parameter. Two refinements matter. The axis is two factors rather than one: of the 333-fold male-mouse-to-human renal clearance gap, the escape fraction carries 50-fold and the six-fold lower human glomerular filtration rate carries 6.5-fold. And humans run a second, enterohepatic loop of comparable completeness, which closes the residual the renal axis leaves and explains why bile-acid sequestrants shorten human half-lives several-fold.

The review also audits the provenance of the numbers regulation uses. The human volume of distribution behind nine adopted clearance factors is a calculation rather than a measurement, reproducible to three figures from its source paper's own supplementary table given an assumed elimination rate drawn from a study in the same two communities; and a widely-inherited plasma free fraction is roughly 164-fold too high because a calculated lower bound was read as a measurement. Eighteen such corrections are documented, one of them to this work's own method.

Finally, half-life is argued to be the wrong endpoint for any structure-activity model, because it carries a body-size term. A dimensionless renal handling ratio is proposed in its place and evaluated on the only dataset that supports the comparison.

1. Introduction: one identity, three questions

Per- and polyfluoroalkyl substances persist in the body for a long time, and how long is the single number that drives their regulation: it converts an external dose into an internal one, and it is what any cross-species extrapolation has to carry. The published values do not agree. Human PFOA half-life estimates span roughly seventeen-fold, from about half a year to eight and a half; the male rat and the female rat differ by about seventy-fold in the same experiment [1, 2]; and the rat and the mouse, two rodents of similar size, differ in a direction that reverses between sexes [3].

Disagreements of that size usually mean a quantity is being compared across things that are not comparable. The way into the problem is that half-life is not an independent property. It is fixed by an identity:

$$t_{1/2} = \ln 2 \cdot V_d / CL$$

where V_d is the apparent volume of distribution - the number relating total body burden to plasma concentration, not an anatomical volume - and CL is clearance, the volume of plasma irreversibly cleared per unit time. Any two of the three terms fix the third. A study that measures serum decay obtains the rate constant directly and never needs a volume; a study that works from mass balance needs a volume and a complete excretion accounting, each uncertain severalfold, and both multiply into the answer. The practical consequence runs through this entire review: most published disagreements are not disagreements about measurements. They are differences in **which two terms were measured and which one was assumed** - and in the most consequential case, the assumed one is the number everybody quotes.

Three questions follow, and this review answers them in order. First, is the species difference a difference in how much is held or in how much is let go - distribution or clearance? Second, what did each reported half-life assume? Third, does exposure level itself change elimination, as saturable kinetics would predict?

2. Evidence base and method

The review rests on 186 full texts read directly, 24 per-paper extractions made from those texts, and a consolidated database of 1,438 rows across four schemas - toxicokinetic parameters, protein binding, transporter kinetics, and adopted regulatory values. Animal serum time courses come from the US EPA's compiled per-record database [4], which digitised or transcribed them from the primary studies - principally the monkey intravenous series [5, 6], the rodent dose series [7, 8, 9] and a low-dose chronic study [10]. Human values come from the primary papers and from agency assessments [11, 12, 13, 14, 15, 16].

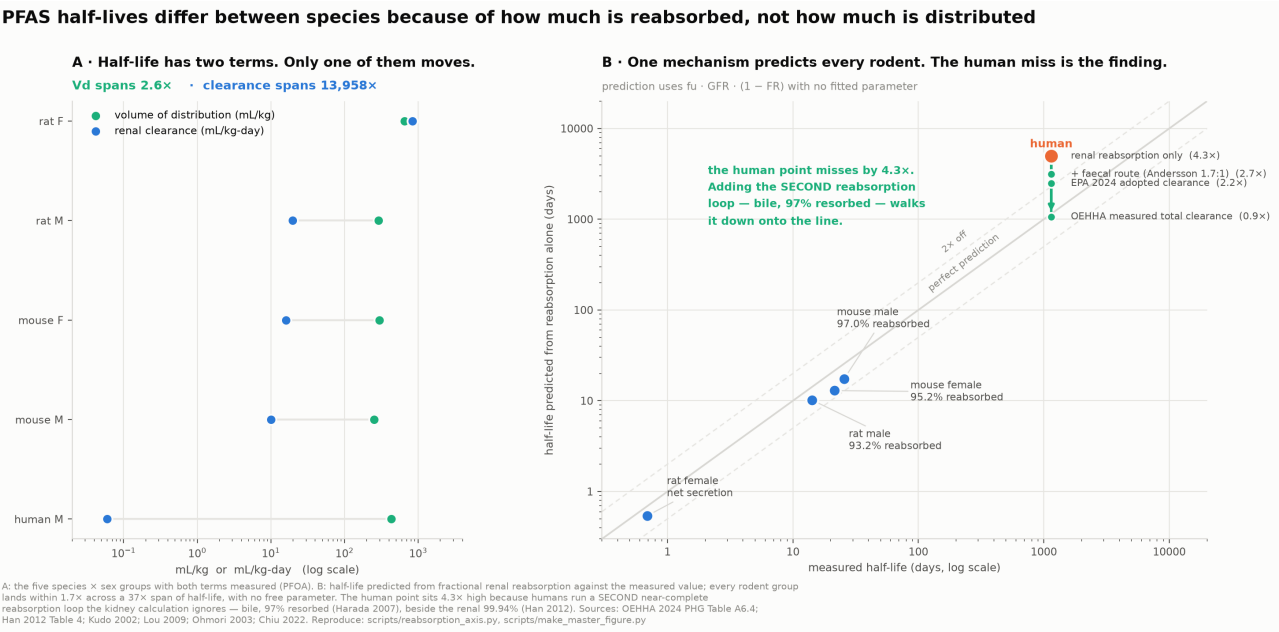
Three rules govern what is in the database, and they are the reason several of this review's conclusions differ from the secondary literature.

Read the table, not the sentence. Where a paper's abstract and its own tables disagree, the table wins and the discrepancy is recorded. Two of the largest corrections in section 11 are of exactly this kind, and so is the correction this review makes to its own earlier work.

Mark what was computed. Every row records whether a value was measured, fitted, assumed, or computed in this work, and carries a PMID or DOI and the table it came from. A clearance derived from an assumed volume is not the same kind of object as a clearance measured from urine, and conflating the two is how the problem in section 9 propagated.

Flag your own biases rather than deleting them. This review refitted the EPA raw curves and found its own fitting procedure biased on biphasic data. The affected column is kept, flagged, and documented, because it remains valid for the within-curve comparisons it is used for - and because deleting it would hide a failure mode that applies to much of the published literature too.

Figure 1



The argument in one pair of panels. Left: across five species-by-sex groups in which both terms were measured, the volume of distribution spans 2.6x while renal clearance spans 13,958x - the variance is in clearance, not distribution (section 3). Right: half-life predicted from fractional renal reabsorption alone, against observation, on log axes with the 1:1 line. Every rodent point falls within 1.7x over a 37x span with no fitted parameter (section 4). The human point sits 4.3x high, and the walk-down beneath it - renal only 4.3x, plus faecal 2.7x, EPA clearance 2.2x, OEHHHA measured clearance 0.9x - is what led to the second reabsorption loop (section 6). Regenerate with scripts/make_master_figure.py.

3. Where the species difference lives: distribution or clearance?

The identity in section 1 makes this testable rather than rhetorical, provided both terms are measured in the same animals. Six datasets satisfy that condition - three species, two compounds, four laboratories, no cross-study comparison required [1, 2, 17, 18, 7, 8].

	V _d male/female	clearance female/male
span across all six datasets	1.33 - 2.18x	0.78 - 44.3x

V_d male/female	clearance female/male
a 1.6× spread	a 56× spread

The distribution ratio is male-higher in every dataset and never leaves a narrow band, which is what one would expect of a quantity set mostly by body composition and plasma protein concentration. The clearance ratio ranges over nearly two orders of magnitude **and changes sign between species**. That asymmetry is the central finding of this review, and everything after it is an attempt to explain the clearance term.

A single recent experiment reproduces the result internally. Argoul and colleagues dosed eleven PFAS as one cocktail in female mice and fitted all of them simultaneously [19]; across those eleven compounds, clearance spans 5,254-fold while steady-state volume of distribution spans 7.7-fold. Same animals, same assay, same model - and the same conclusion as the cross-study comparison.

Taken from primary sources rather than compilations, the species gap is also concentrated in one sex. Female rat divided by female mouse PFOA clearance is 373-496×, against 7.0× in males [1, 20]. Tatum-Gibbs and colleagues replicate the whole structure in a second compound with strains matched: a rat sex ratio of 21.9× against a mouse ratio near unity [3]. Any account of 'the species difference' that does not mention sex is describing an average over two quite different animals.

4. The mechanism: fractional renal reabsorption

PFAS are filtered freely at the glomerulus to the extent they are unbound, and then substantially recovered from the tubular filtrate. Renal clearance is therefore what escapes:

$$CL_{renal} = f_u \cdot GFR \cdot (1 - FR)$$

where f_u is the unbound fraction in plasma and FR the fraction reabsorbed. Han and colleagues assembled this across species [21]; the table below is read from that paper's Table 4 at source rather than from any adaptation of it.

species	sex	GFR (L/d/kg)	renal CL (mL/d/kg)	reabsorbed
human	both	2.57	0.03	99.94%
mouse	female	16.7	16	95.2%
mouse	male	16.7	10	97.0%
rat	female	14.4	666	net secretion
rat	male	14.4	18.2	93.7%
Japanese macaque	female	8.5	32	81.2%
Japanese macaque	male	8.5	15	91.2%
dog	female	5.3	50.8	52%
dog	male	5.3	43	59%
rabbit	female	4	670	net secretion
rabbit	male	4	640	net secretion

Read at source into `db/primary_2026/han2012_table4_reabsorption_axis.csv`. Two values differ from a widely-cited adaptation of this table; see section 11.

The ordering is the half-life ordering. More than that, the axis *predicts*: substituting each species' own GFR and unbound fraction reproduces rodent half-lives within 1.7× over a 37× span with no fitted parameter, and an entirely separate dataset recomputes mouse PFOA reabsorption at 95.9% [19], inside the 95.2-97.0% range already in use. For a field in which half-lives disagree seventeen-fold, a parameter-free prediction good to 1.7× is a strong result.

Elimination hands off from kidney to gut as the chain lengthens

The axis is renal, and for the shorter carboxylates that is nearly the whole story: PFHpA leaves the male rat 92% in urine over five days. By PFNA that falls to 2.0% and by PFDA to 0.2%, with faeces becoming the major route [22]. So a renal clearance is close to a total clearance at C7 and is a small minority of it at C10, and any human accounting that assigns one route fraction across compounds is wrong in a predictable direction. The handoff is also sex-dependent: female rats excrete 51% of a PFNA dose in urine where males excrete 2.0% [22], which is the same sex difference as section 3 seen through the route split rather than the rate.

The axis has two factors, not one

It is tempting to read the table as 'humans reabsorb more', and this review initially did. The absolute numbers say otherwise: humans recover 51 mL/d/kg of filtrate, against 270 in the male rat and 318-324 in the mouse. Humans reabsorb *less* in absolute terms and still clear far more slowly, because they filter far less to begin with. Decomposing the 333× male-mouse-to-human renal clearance gap gives 50× from the escape fraction and 6.5× from the six-fold lower human GFR - 68% and 32% on a log scale. Reabsorption is the larger term, which is why the single-axis framing works at all; but a third of the difference is filtration rate, and no transporter story explains that third.

5. Transporter identity, and why the mechanism does not transfer

If reabsorption is the controlling step, some apical transporter must perform it, and in the rat the sex difference constrains which one. A candidate has to satisfy two conditions simultaneously: it must be sex-divergent in the right direction, and it must actually carry PFOA. Those two conditions are jointly much more restrictive than either alone.

transporter	sex-divergent?	transports PFOA?	verdict
Oatp1a1	yes, 23× male-predominant, androgen-induced	yes, C8-C10	the only candidate satisfying both
Oat2	yes, strongly	no - three negative reports, two species	excluded by transport
Oat1 / Oat3	no	yes	excluded by regulation

The transport evidence is from direct uptake assays [23, 24, 25] and the regulation evidence from expression studies under hormonal manipulation [1, 26, 27, 28]; the organic anion transporting polypeptides as a family have been shown to contribute to PFAA disposition in both humans and rats [29]. The 23× figure is from the primary source; a frequently quoted '2.5-fold' belongs to a different transporter (see section 11).

The surprise is in the human

The reabsorbed *fraction* transfers across species - that is what section 4 shows. The *mechanism* does not. OATP1A2, the closest human orthologue of rat Oatp1a1, does not mediate saturable PFOA uptake at all [30]. Human apical reabsorption instead runs through OAT4 and URAT1 [30, 31], neither of which is androgen-regulated - which is consistent with the absence in humans of anything like the rat's seventyfold sex difference. Extrapolating a rat mechanism to humans and extrapolating a rat reabsorbed fraction to humans are therefore very different acts, and only the second is supported.

6. The second loop: enterohepatic recirculation

The renal axis leaves a residual. Predicted human half-life comes out about 4.3× short of observation - visible as the single off-line point in Figure 1. Three candidate explanations exist: the human reabsorbed fraction is underestimated, the human clearance figure is wrong, or a second elimination route is being reabsorbed too.

The third is correct, and the number was already in an agency document. Harada and colleagues sampled serum and bile from four gallstone-surgery patients and measured a biliary resorption rate of 0.97 [32]. Bile PFOS (27.9 ng/mL) actually *exceeds* serum (23.2), so bile is a genuine excretion route - but 97% of what is secreted is recovered from the gut.

loop	fraction reabsorbed	source
renal (PFOA)	99.94%	Han 2012 Table 4 [21]
biliary (PFOS)	97%	Harada 2007, n = 4 [32]

Humans therefore run two near-complete reabsorption loops rather than one. Both are near unity, both lengthen the half-life, and - the testable part - **both are interruptible**. If a 97% resorption loop is blocked, elimination should accelerate several fold, and it does. In a treatment programme using cholestyramine and plasma donation, apparent PFOS half-life was 1.2 y in nineteen treated participants against 7.3 y in nine observed ones, and PFHxS 2.5 y against 9.4 y [33]. Supporting evidence comes from a cholestyramine series [34] and a cross-over trial reporting 63% lowering in twelve weeks against 3% in controls [35]. Plasma donation alone, which removes burden without touching either loop, produces a much smaller effect [36].

The loop also settles an apparent contradiction in the human excretion literature. One study measuring faeces under ongoing intake finds faecal elimination dominant for PFOS [37]; another, following a single labelled dose, barely detects it [38]. At 97% resorption, **gross** biliary flux is large while **net** faecal elimination is small, so the two studies measured

different quantities and both are right. The caveat is that the constant doing this work rests on four patients.

7. Plasma protein binding, and a conflict that was not one

The unbound fraction f_u enters the reabsorption equation directly, so its value propagates into everything above. The literature appeared to contain a hundredfold disagreement: a long-standing result reports PFOA 'over 90% bound' to plasma protein [39], implying f_u near 0.1, while recent measurements at environmentally realistic ligand:protein ratios give 0.00061 [40].

Reading both papers at source dissolves the conflict. The '>90%' is not a measurement but a *calculation* from a dissociation constant and an albumin concentration, and as a lower bound it is satisfied by 0.00061 as comfortably as by 0.1. The underlying difference in dissociation constant is a titration artefact: the older work titrated 50-60 μM albumin with 0.1-3 mM PFOA, a ligand:protein ratio of 1.7:1 to 60:1, against 0.004:1 or below in the newer work - while human serum sits at 10^{-5} to 10^{-3} :1. Near-saturating ratios understate affinity, and structural work on the albumin complex is consistent with multiple sites of differing affinity [41].

The defect was in the inheritance rather than in either measurement. Physiologically based models read '>90% bound' as 'about 90% bound' and adopted a free fraction roughly **164× too high**. Section 10 shows how far that single substitution propagates.

8. Does exposure level change elimination?

If reabsorptive transport saturates, elimination should accelerate at high burden, half-life should fall with dose, and a single clearance factor could not be transferred between a contaminated community and the general population. The evidence is weak but not null, and it splits informatively by the level at which the comparison is made.

comparison	direction
within a person over time	supports dose dependence
within a species across doses (rat refits)	supports, weakly: slope about +0.11
between people in a cohort	contradicts - the lowest-exposure tertile declines fastest
between water districts	slope above the mechanistic ceiling of 1, i.e. bias

Within-unit comparisons support concentration dependence and between-unit ones do not, which is the signature of a between-unit confounder - and the source study names it: age [42]. Re-running that paper's own variance decomposition, age carries 2-13× the partial R^2 of initial PFAS concentration, and the exposure-tertile effect is 25-37% the size of the age effect. Older people eliminate more slowly and have higher accumulated burdens, so the between-person association is a positive saturation effect minus a larger negative age effect.

Across volume of distribution the slopes run -0.21 to +0.25 with inconsistent sign, i.e. no relationship. The one clean exception runs the other way: the female rat shows a slope of +0.21 over a 3,200× dose range against -0.02 in the male, which is what saturable *secretion* looks like - consistent with the female rat sitting at the secretory end of the axis in section 4.

Whether human exposures reach saturation is unresolved and is taken up in section 13.

9. The human volume of distribution

Nine of the 42 adopted regulatory clearance factors catalogued here are computed from a single human volume of distribution, 170 mL/kg [43]. That number is not a measurement.

Its source paper's supplementary table is headed '*Input data for the calibration of the Vd parameter*' with a final column headed '*calculated Vd*', and its published values reproduce to three figures from intake, serum concentration and an assumed elimination rate. The assumed rate was taken from a study that measured half-life in **the same two communities** [44], which is what makes the construction circular rather than merely indirect.

The consequence is sharper than circularity, and it is asymmetric. Forming a clearance from the identity makes the half-life cancel exactly:

half-life assumed	V_d that follows	derived clearance
2.3 y (the value used)	168 mL/kg	0.132-0.138 mL/kg-day
3.8 y	277 mL/kg	0.132-0.138 mL/kg-day (<i>unchanged</i>)

Clearance reduces to dose divided by serum concentration and is robust; the volume does not cancel and is proportional to whatever half-life was assumed. So adopting '170' alongside a different half-life silently contradicts the data it came from. The arithmetic is reproduced from the paper's own supplementary table to within 1.000-1.003× in scripts/thompson_vd_circularity.py. The paper's own corrigendum is an instance of exactly this error, corrected by a factor of 0.6 - the ratio of two rate constants.

Independent measurements do not support the adopted value in either direction: direct estimates cluster at 74, 121 and 113-199 mL/kg [37, 38, 36], all *below* the assigned range, while the one population pharmacokinetic fit gives 430 mL/kg [45], far above it. A mass-balance half-life anchored on 170 mL/kg [46] moves to within 5% of the population fit when the fitted volume is substituted instead, which suggests the seventeen-fold human controversy is largely one assumed constant.

10. Toward structure-activity: the renal handling ratio

A long-term aim of this work is to relate PFAS toxicokinetics to molecular structure. Half-life cannot be that endpoint. Because it is $\ln 2 \cdot V_d / CL$, it carries a glomerular filtration term that differs 6.5× between mouse and human with no change in chemistry whatever; a structural model fitted to it is being asked to absorb body size, and it cannot.

The replacement proposed here is dimensionless - measured renal clearance divided by the clearance free filtration alone would produce:

$$R = CL_{renal} / (f_u \cdot GFR)$$

R below 1 is net reabsorption, R above 1 net secretion, R = 1 pure filtration. It divides out GFR, so species of different size become comparable, and divides out f_u , so the binding step is not counted twice; what remains is the transport step, which is the part a structural model could plausibly learn. It is the axis of section 4 re-expressed, since $R = 1 - FR$, on a scale that neither crowds against a ceiling at 1 nor runs unboundedly negative under secretion.

Only one dataset supports the comparison: nine compounds with both renal clearance and unbound fraction measured in one experiment, one species, one laboratory [19]. R spans 875× across them.

compound	F-carbons	head group	ether O	f_u %	$\log_{10} R$	handling
PFDA	9	-COOH	0	0.42	-2.509	reabsorbed
PFHxS	6	-SO ₃ H	0	1.3	-2.344	reabsorbed
PFNA	8	-COOH	0	0.35	-1.766	reabsorbed
PFOA	7	-COOH	0	0.87	-1.384	reabsorbed
PFOS	8	-SO ₃ H	0	0.25	-1.305	reabsorbed
PFBA	3	-COOH	0	77.0	-0.993	reabsorbed
GenX	5	-COOH, ether	1	26.0	-0.985	reabsorbed
PFO2OA	5	-COOH, ether	2	10.0	0.186	SECRETED
PFHxA	5	-COOH	0	25.0	0.433	SECRETED

Built by scripts/qsar_endpoint_table.py into db/qsar/qsar_endpoint_table.csv. 'F-carbons' counts carbons bearing fluorine, which excludes a carboxylate's acid carbon and includes every carbon of a sulfonate - a definition chosen so that PFNA and PFOS come out the same size, which the data can then falsify.

Three results constrain any structure-activity model built on this endpoint. **Chain length alone does not order it:** within the carboxylates the series is non-monotonic and PFHxA crosses into net secretion, so a model using carbon number as its sole descriptor is already falsified on nine compounds. **The head group carries about threefold at matched chain length** - PFNA against PFOS, both with eight fluorinated carbons, differ 2.9×, so the equivalence the descriptor assumed is a good approximation but not free. **Ether oxygens move the endpoint 1.4 log units at constant chain length, and non-monotonically:** PFHxA with none is secreted, GenX with one is reabsorbed, PFO2OA with two is secreted. The replacement chemicals sit on both sides of the divide, which is a regulatory observation as much as a chemical one.

A sensitivity that bears on the species question

R is linear in $1/f_u$, so it inherits the uncertainty of section 7 in full. For human PFOA, with renal clearance and GFR both fixed, the three free fractions in circulation give three different answers:

f_u	source	$\log_{10} R$	gap to male mouse
0.10	models reading '>90% bound' as 'about 90%'	-3.93	354×
0.02	the stated assumption in [21]	-3.23	71×
0.00061	measured at physiological ligand:protein [40]	-1.72	2.2×

Under the assumed value the mouse-to-human difference in the transport step is about seventyfold and the species gap is transport biology. Under the measured value it is about twofold, and the species gap is almost entirely **binding** - which would mean the transporter literature of section 5 is explaining a quantity that barely differs between the species. This is stated as a sensitivity rather than a result, because the mouse and human free fractions come from different methods at different ligand:protein ratios, which is precisely the artefact section 7 diagnoses. It is also why the experiment proposed in section 13 is the one worth doing first.

11. What this review corrects in the published record

Each entry is traced to the table or figure that contradicts it; the full list of eighteen is section 8 of the underlying review.

source	correction
US EPA [11]	the rat Oatp1a1 male/female ratio quoted as '2.5-fold' is a <i>different transporter's</i> number (OAT-K); the primary value is 23× [1]
OEHHA [14]	its adaptation of the reabsorption table altered two values (human 99.94 to 99.8, male rat 93.7 to 93.2) relative to the source [21]
Provenance	the $f_u = 0.02$ behind the axis is the source paper's assumption, not the agency's, and that paper calls its own values 'rough estimates' [21]
Andersson 2025 [37]	transposes the PFOA and PFOS volumes of [43] (it is PFOA 170, PFOS 230); an earlier claim in this work that [46] did so was itself wrong and is withdrawn
Cheng 2006 [47]	its abstract calls renal Oatp1a1 'female-predominant', then reports androgens <i>increase</i> it and concludes androgens are the exclusive cause - self-contradictory, and against [26, 27]
Metabolic inertness	'PFAS are metabolically inert' fails for 6:2 Cl-PFESA, which is biotransformed [48, 49]
This review	its own refits of the EPA raw curves compress sex ratios about 13× , as below

The self-correction matters most

This work refitted terminal slopes directly from the EPA raw curves [4], selecting the window with the best adjusted R^2 . On a biphasic curve that criterion selects the shallow terminal tail. Tested against a published value for the same experiment, the fit returns **1.44 d against 0.08 d** for the female rat and compresses a **71× sex ratio to 5.6×** [1]. The bias runs one way, so every conclusion above survives and several strengthen - but that column must not be read as comparable to published half-lives. It carries a flag and is kept rather than deleted, because it remains valid for within-curve comparison and because the same failure mode applies wherever a terminal slope is fitted without inspecting the window.

12. Coverage: how much of the field is empty

A review that only reports what is known overstates the state of the field, so the gap was measured. Crossing 39 chemicals by 11 species by 4 parameters gives 1,716 cells.

	cells
no data at all	1,466 (85%)
resting on a single study	94
PFOS, the best-covered compound in the world	29 of 44 filled
by species: human / rat / mouse / monkey / dog	71 / 61 / 45 / 28 / 2

Mining four further compilations [15, 16, 50, 51] moved coverage from 143 to 156 cells, so the hole is real rather than a search artefact. The dog column deserves particular notice: it is the species that sits furthest down the reabsorption axis

among the mammals with data, and it rests on two cells, one of which traces to a book chapter that could not be obtained.

13. Open questions, and what would close them

1. The unbound fraction, measured one way across species

This is the cheapest decisive experiment the review identifies, and it is first because two separate questions turn on it. Every value of R in section 10 is proportional to f_u , and the mouse and human numbers currently come from different methods at different ligand:protein ratios. Measuring them by one method at physiological ratio decides whether the species difference is binding or transport - a 33× swing in the answer - and simultaneously supplies the denominator any structure-activity model needs. It requires plasma, a dialysis or ultrafiltration rig and a mass spectrometer. No animals.

2. A measured human transport-affinity constant

Whether reabsorptive transport saturates at real human exposures depends on which parameter family one believes. Six independent in vitro half-saturation constants for PFOA against human transporters span 47 to 310 μM , from two laboratories, agreeing within a factor of seven [30, 25, 31]. The transport-affinity constant fitted inside physiologically based models is 0.133 μM - **354 to 2,336× lower than anything ever measured in a cell**. The in vitro values put human serum far below half-saturation; the fitted value puts three of four human populations at or above it, which would make clearance dose-dependent in contaminated communities and mean a single clearance factor cannot transfer between exposure settings. The weight of evidence has moved onto the in vitro side - the fitted values are fitted to plasma curves rather than measured, the mouse row carries standard errors exceeding its estimates, and the dose-response evidence of section 8 agrees with the in vitro answer - but nobody has measured a human value, and one measurement would close it outright.

3. Whether the biliary resorption constant replicates

The 0.97 of section 6 is now load-bearing for the human limb, and it rests on four surgical patients in a single study reached through an agency document [32]. A replication in any species under a stated protocol would be worth more than its cost.

4. The mouse renal Oatp1a1 sex ratio

The rat value is 23× [1]; the mouse equivalent is not established, and the mouse sex difference runs the opposite way to the rat's [3]. One quantitative PCR experiment would discriminate between the live hypotheses - though if the answer to question 1 is that the species gap is binding, this question loses most of its force, which is why it is fourth rather than first.

5. Unobtained sources

Three documents would materially change specific numbers and could not be obtained: the dog toxicokinetic chapter underlying the dog column, an unpublished contract report on protein binding, and the primary rat dose series [9], which supplies roughly 70% of the rat PFOA observations used here only through its digitisation by [4].

14. Conclusions

The species difference in PFAS elimination is a clearance difference, not a distribution difference, and it is concentrated in females. One mechanistic axis - the fraction of filtered compound recovered from the tubule - orders every species for which both terms have been measured, and predicts rodent half-lives within 1.7× with no fitted parameter. That axis is two factors rather than one, roughly two thirds reabsorption and one third glomerular filtration rate, and in humans it is joined by a second, enterohepatic loop of comparable completeness whose interruption shortens human half-lives several-fold.

The reabsorbed fraction transfers between species; the protein performing the reabsorption does not. That distinction matters for how animal data should be read across to humans: the quantitative axis is extrapolable, the mechanism is not.

Against that, two of the most widely inherited human parameters do not mean what their users take them to mean. The volume of distribution behind nine adopted clearance factors is a calculation whose assumed half-life came from the same communities it was calibrated on, and a plasma free fraction in general use is roughly 164× too high because a calculated lower bound was read as a measurement. Neither is a disputed measurement; both are provenance failures, and both are fixable by reading the source tables.

Finally, half-life should not be the endpoint of a structure-activity model, because it carries body size. The dimensionless renal handling ratio proposed here removes both the filtration and the binding terms, spans 875× across the nine compounds that support the comparison, and shows immediately that chain length alone will not do: the head group carries threefold at matched length and ether substitution moves the endpoint 1.4 log units non-monotonically, placing the replacement chemicals on both sides of the reabsorption-secretion divide. Extending that comparison to a second species, with unbound fraction measured the same way in both, is the single step that would turn this argument into a model.

Data and code availability

All extractions, the consolidated 1,438-row database, the 34 analysis scripts that regenerate every figure and table above, and the full review are at github.com/sdey17/pfas_tk. Every database row carries a PMID or DOI, the table it was read from, and whether its value was measured, fitted, assumed or computed in this work. Animal serum time courses are redistributed from the EPA compilation [4] under its MIT licence.

References

Numbered by order of first citation. Entries marked as held through a secondary source were not obtained in the original; the review does not quote them as if they had been.

1. Kudo N, Katakura M, Sato Y, Kawashima Y (2002). Sex hormone-regulated renal transport of perfluorooctanoic acid. *Chemico-Biological Interactions* 139(3):301-316. doi:10.1016/s0009-2797(02)00006-6
2. Ohmori K, Kudo N, Katayama K, Kawashima Y (2003). Comparison of the toxicokinetics between perfluorocarboxylic acids with different carbon chain length. *Toxicology* 184(2-3):135-140. doi:10.1016/s0300-483x(02)00573-5
3. Tatum-Gibbs K, Wambaugh JF, Das KP, Zehr RD, Strynar MJ, Lindstrom AB, Delinsky A, Lau C (2011). Comparative pharmacokinetics of perfluorononanoic acid in rat and mouse. *Toxicology* 281(1-3):48-55. doi:10.1016/j.tox.2011.01.003
4. Zurlinden TJ, Dzierlenga MW, Kapraun DF, Ring C, Bernstein AS, Schlosser PM, Morozov V (2025). Estimation of species- and sex-specific PFAS pharmacokinetics in mice, rats, and non-human primates using a Bayesian hierarchical methodology. *Toxicology and Applied Pharmacology* 499:117336. doi:10.1016/j.taap.2025.117336. Data: github.com/USEPA/CPHEA-Animal-PFAS-PK
5. Butenhoff JL, Kennedy GL, Hinderliter PM, Lieder PH, Jung R, Hansen KJ, Gorman GS, Noker PE, Thomford PJ (2004). Pharmacokinetics of perfluorooctanoate in cynomolgus monkeys. *Toxicological Sciences* 82(2):394-406. doi:10.1093/toxsci/kfh302
6. Chang S-C, Noker PE, Gorman GS, Gibson SJ, Hart JA, Ehresman DJ, Butenhoff JL (2012). Comparative pharmacokinetics of perfluorooctanesulfonate (PFOS) in rats, mice, and monkeys. *Reproductive Toxicology* 33(4):428-440. doi:10.1016/j.reprotox.2011.07.002
7. Dzierlenga AL, Robinson VG, Waidyanatha S, DeVito MJ, Eifrid MA, Gibbs ST, Granville CA, Blystone CR (2020). Toxicokinetics of PFHxA, PFOA and PFDA in male and female Hsd:Sprague Dawley SD rats following intravenous or gavage administration. *Xenobiotica* 50(6):722-732. doi:10.1080/00498254.2019.1683776
8. Huang MC, Dzierlenga AL, Robinson VG, Waidyanatha S, DeVito MJ, Eifrid MA, Granville CA, Gibbs ST, Blystone CR (2019). Toxicokinetics of perfluorobutane sulfonate, perfluorohexane-1-sulphonic acid, and perfluorooctane sulfonic acid in male and female Hsd:Sprague Dawley SD rats after intravenous and gavage administration. *Toxicology Reports* 6:645-655. doi:10.1016/j.toxrep.2019.06.016, with its corrigendum, *Toxicology Reports* 8:365. doi:10.1016/j.toxrep.2021.02.001
9. Kemper RA (2003). Perfluorooctanoic acid: toxicokinetics in the rat. Unpublished report, DuPont-7473; US EPA public docket AR-226-1499. Haskell Laboratory, E.I. du Pont de Nemours. (Not obtained; its data are used as digitised by the EPA database [zurlinden2025].)
10. Iwabuchi K, Senzaki N, Mazawa D, Sato I, Hara M, Ueda F, Liu W, Tsuda S (2017). Tissue toxicokinetics of perfluoro compounds with single and chronic low doses in male rats. *Journal of Toxicological Sciences* 42(3):301-317. doi:10.2131/jts.42.301
11. US EPA (2024). *Final Human Health Toxicity Assessment for Perfluorooctanoic Acid (PFOA)*, EPA-815R24006, and its Appendix (Table B-26).
12. US EPA (2024). *Final Human Health Toxicity Assessment for Perfluorooctane Sulfonic Acid (PFOS)*.
13. US EPA (2023). *IRIS Toxicological Review of Perfluorohexanoic Acid (PFHxA)*.
14. California OEHHA (2024). *Public Health Goals for PFOA and PFOS in Drinking Water*, Appendix Tables A6.3, A6.4 and 4.8.1.
15. ATSDR (2021). *Toxicological Profile for Perfluoroalkyls*, Tables 3-5 and 3-6.
16. EFSA CONTAM Panel (2020). Risk to human health related to the presence of perfluoroalkyl substances in food. *EFSA Journal* 18(9):6223. doi:10.2903/j.efsa.2020.6223
17. Sundström M, Chang S-C, Noker PE, Gorman GS, Hart JA, Ehresman DJ, Bergman Å, Butenhoff JL (2012). Comparative pharmacokinetics of perfluorohexanesulfonate (PFHxS) in rats, mice, and monkeys. *Reproductive Toxicology* 33(4):441-451. doi:10.1016/j.reprotox.2011.07.004
18. Kim S-J, Heo S-H, Lee D-S, Hwang IG, Lee Y-B, Cho H-Y (2016). Gender differences in pharmacokinetics and tissue distribution of 3 perfluoroalkyl and polyfluoroalkyl substances in rats. *Food and Chemical Toxicology* 97:243-255. doi:10.1016/j.fct.2016.09.017

19. Argoul CML, Toutain P-L, Picard-Hagen N, Mselli-Lakhal L, Dauwe Y, Roques BB, Lacroix MZ, Gayrard V (2026). Nonlinear mixed-effects modeling of the intravenous and oral kinetics of eleven perfluoroalkyl substances in female mice. *Environmental Research* 303:124802. doi:10.1016/j.envres.2026.124802
20. Lou I, Wambaugh JF, Lau C, Hanson RG, Lindstrom AB, Strynar MJ, Zehr RD, Setzer RW, Barton HA (2009). Modeling single and repeated dose pharmacokinetics of PFOA in mice. *Toxicological Sciences* 107(2):331-341. doi:10.1093/toxsci/kfn234
21. Han X, Nabb DL, Russell MH, Kennedy GL, Rickard RW (2012). Renal elimination of perfluorocarboxylates (PFCAs). *Chemical Research in Toxicology* 25(1):35-46. doi:10.1021/tx200363w
22. Kudo N, Suzuki E, Katakura M, Ohmori K, Noshiro R, Kawashima Y (2001). Comparison of the elimination between perfluorinated fatty acids with different carbon chain length in rats. *Chemico-Biological Interactions* 134(2):203-216. doi:10.1016/s0009-2797(01)00155-7
23. Yang C-H, Glover KP, Han X (2009). Organic anion transporting polypeptide (Oatp) 1a1-mediated perfluorooctanoate transport and evidence for a renal reabsorption mechanism of Oatp1a1 in renal elimination of perfluorocarboxylates in rats. *Toxicology Letters* 190(2):163-171. doi:10.1016/j.toxlet.2009.07.011
24. Weaver YM, Ehresman DJ, Butenhoff JL, Hagenbuch B (2010). Roles of rat renal organic anion transporters in transporting perfluorinated carboxylates with different chain lengths. *Toxicological Sciences* 113(2):305-314. doi:10.1093/toxsci/kfp275
25. Louisse J, Pedroni L, van den Heuvel JJMW, Rijkers D, Leenders L, Noorlander A, Punt A, Russel FGM, Koenderink JB, et al. (2024). In vitro and in silico characterization of the transport of selected perfluoroalkyl carboxylic acids and perfluoroalkyl sulfonic acids by human organic anion transporter 1 (OAT1), OAT2 and OAT3. *Toxicology* 509:153961. doi:10.1016/j.tox.2024.153961
26. Cheng X, Maher J, Chen C, Klaassen CD (2005). Tissue distribution and ontogeny of mouse organic anion transporting polypeptides (Oatps). *Drug Metabolism and Disposition* 33(7):1062-1073. doi:10.1124/dmd.105.003640
27. Cheng X, Klaassen CD (2009). Tissue distribution, ontogeny, and hormonal regulation of xenobiotic transporters in mouse kidneys. *Drug Metabolism and Disposition* 37(11):2178-2185. doi:10.1124/dmd.109.027177
28. Buist SCN, Klaassen CD (2004). Rat and mouse differences in gender-predominant expression of organic anion transporter (Oat1-3; Slc22a6-8) mRNA levels. *Drug Metabolism and Disposition* 32(6):620-625. doi:10.1124/dmd.32.6.620
29. Zhao W, Zitzow JD, Weaver Y, Ehresman DJ, Chang SC, Butenhoff JL, Hagenbuch B (2017). Organic anion transporting polypeptides contribute to the disposition of perfluoroalkyl acids in humans and rats. *Toxicological Sciences* 156(1):84-95. doi:10.1093/toxsci/kfw236
30. Yang C-H, Glover KP, Han X (2010). Characterization of cellular uptake of perfluorooctanoate via organic anion-transporting polypeptide 1A2, organic anion transporter 4, and urate transporter 1 for their potential roles in mediating human renal reabsorption of perfluorocarboxylates. *Toxicological Sciences* 117(2):294-302. doi:10.1093/toxsci/kfq219
31. Louisse J, et al. (2023). Perfluoroalkyl substances (PFASs) are substrates of the renal human organic anion transporter 4 (OAT4). *Archives of Toxicology*. doi:10.1007/s00204-022-03428-6
32. Harada K, et al. (2007). Biliary excretion and enterohepatic recirculation of perfluorooctane sulfonate in humans. Values used here are as tabulated by US EPA (2016), *Health Effects Support Document for Perfluorooctane Sulfonate*; the original was not obtained.
33. Delaere I, Harris K, Gaskin S, Tefera Y, Mitchell K, Springer D, Mills S (2025). Changes in serum perfluorooctane sulfonic acid and perfluorohexane sulfonic acid concentrations in firefighters accessing a voluntary PFAS reduction treatment program. *Environment International* 202:109609. doi:10.1016/j.envint.2025.109609
34. Genuis SJ, et al. (2010). Human elimination of perfluorinated compounds under cholestyramine administration. (Cited via [andersson2025] and the EPA assessment; the original was not obtained.)
35. Møller S, et al. (2024). Cholestyramine cross-over trial for PFAS lowering. *Environment International*. doi:10.1016/j.envint.2024.108471 (cited via [andersson2025])
36. Gasiorowski R, Forbes MK, Silver G, Krastev Y, Hamdorf B, Lewis B, et al. (2022). Effect of plasma and blood donations on levels of perfluoroalkyl and polyfluoroalkyl substances in firefighters in Australia: a randomized clinical trial. *JAMA Network Open* 5(4):e226257. doi:10.1001/jamanetworkopen.2022.6257
37. Andersson AG, et al. (2025). The relative importance of fecal and urinary excretion of perfluorooctane sulfonic acid and perfluorooctanoic acid after high exposure: an observational study in Ronneby, Sweden. *Environmental Research* 285:122487. doi:10.1016/j.envres.2025.122487
38. Abraham K, Mertens H, Richter L, Mielke H, et al. (2024). Kinetics of 15 PFAS after a single oral dose in one adult volunteer: terminal half-lives, clearances and derived volumes of distribution. *Environment International*. doi:10.1016/j.envint.2024.109047
39. Han X, Snow TA, Kemper RA, Jepson GW (2003). Binding of perfluorooctanoic acid to rat and human plasma proteins. *Chemical Research in Toxicology* 16(6):775-781. doi:10.1021/tx034005w
40. Fischer FC, et al. (2024). Protein binding of PFAS measured by solid-phase microextraction at environmentally relevant PFAS:protein ratios. *Environmental Science & Technology*. doi:10.1021/acs.est.3c07415, and Fischer FC, et al. (2025). doi:10.1021/acs.est.5c05473
41. Maso L, Trande M, Liberi S, Moro G, Daems E, Linciano S, et al. (2021). Unveiling the binding mode of perfluorooctanoic acid to human serum albumin. *Protein Science* 30(4):830-841. doi:10.1002/pro.4036
42. Li Y, Andersson A, Xu Y, Pineda D, Nilsson CA, Lindh CH, Jakobsson K, Fletcher T (2022). Determinants of serum half-lives for perfluoroalkyl substances after end of exposure to contaminated drinking water, Ronneby cohort. (Held as a structured abstract; full text paywalled.)

43. Thompson J, Lorber M, Toms L-ML, Kato K, Calafat AM, Mueller JF (2010). Use of simple pharmacokinetic modeling to characterize exposure of Australians to perfluorooctanoic acid and perfluorooctane sulfonic acid. *Environment International* 36(4):390-397. doi:10.1016/j.envint.2010.02.008, with its corrigendum, *Environment International* 36(6):652. doi:10.1016/j.envint.2010.05.008
44. Bartell SM, Calafat AM, Lyu C, Kato K, Ryan PB, Steenland K (2010). Rate of decline in serum PFOA concentrations after granular activated carbon filtration at two public water systems in Ohio and West Virginia. *Environmental Health Perspectives* 118(2):222-228. doi:10.1289/ehp.0901252
45. Chiu WA, et al. (2022). Bayesian estimation of human population toxicokinetics of PFOA, PFOS, PFHxS and PFNA from studies of contaminated drinking water. *Environmental Health Perspectives* 130(12):127001. doi:10.1289/EHP10103
46. Zhang Y, Beesoon S, Zhu L, Martin JW (2013). Biomonitoring of perfluoroalkyl acids in human urine and estimates of biological half-life. *Environmental Science & Technology* 47(18):10619-10627. doi:10.1021/es401905e
47. Cheng X, Maher J, Lu H, Klaassen CD (2006). Endocrine regulation of gender-divergent mouse organic anion-transporting polypeptide (Oatp) expression. *Molecular Pharmacology* 70(4):1291-1297. doi:10.1124/mol.106.025122 (abstract only; full text paywalled)
48. Yi S, et al. (2022). Biotransformation of 6:2 chlorinated polyfluoroalkyl ether sulfonate in the rat. (Held as a structured abstract; see db/primary_2026/yi2022_clpfesa_rat.csv.)
49. Shi Y, Vestergren R, Xu L, Zhou Z, Li C, Liang Y, Cai Y (2016). Human exposure and elimination kinetics of chlorinated polyfluoroalkyl ether sulfonic acids (Cl-PFESAs). *Environmental Science & Technology* 50(5):2396-2404. doi:10.1021/acs.est.5b05849
50. US EPA (2025). *IRIS Toxicological Review of Perfluorohexanesulfonic Acid (PFHxS)*, Table 3-3.
51. New Jersey Drinking Water Quality Institute (2017-2018). Health-based maximum contaminant level support documents for PFOA, PFOS and PFNA.